

Price Configurator Back Office - Strategy & Phased Delivery

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Repositories: price-configurator (frontend), price-configurator-back-office (new), kitchen-quiz-backoffice (reference)

1. Executive summary

Today, kitchen catalog data, price tables, and brand configuration live as **hardcoded JavaScript** under `src/brands/{brand}/` and `config/{brand}/messages.json` in the Zeus React **price-configurator** app. Business users cannot change prices without a developer deploy.

The strategy is to introduce **price-configurator-back-office**: an ASP.NET Core CMS service (same stack as **kitchen-quiz-backoffice**) that becomes the **source of truth** for all configurator content. The public website loads **published** configuration via a read API; price **calculation** stays in the frontend.

Goal	Approach
Editable prices & catalog	Nobia.CmsToolkit + Cosmos DB
Safe go-live	Draft -> Publish, with scheduled publish
Minimal frontend risk	API returns same shape the UI uses today
Rollout	Magnet first, then five other brands

2. Problem statement

Current state

Concern	Location today
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Sections, cards, tooltips	src/brands/{brand}/index.js (Magnet ~1000 lines)
Price lookup tables	src/brands/{brand}/prices.js
Cloudinary image IDs	src/brands/{brand}/cloudinaryImages.js
UI copy / labels	config/{brand}/messages.json
Price calculation rules	src/components/cards/utils.js
Quote submit	GraphQL -> price-configurator-api (unchanged)

Brands: magnet, marbodal, invita, sigdal, norema, novart.

There is **no configuration API** at runtime; data is bundled at build time.

Desired state

- Editors use **price-configurator-back-office** to manage catalog, prices, images, and (eventually) all copy.
 - **price-configurator** fetches published config from the backoffice API on load.
 - Price changes go live after **publish** (immediately or on a **schedule**).
 - No production dependency on redeploying the React app for data changes.
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3. Architecture

3.1 System context

Data flow (top to bottom):

1. **Editors** - Azure AD, per-brand admin groups
2. **price-configurator-back-office** - CmsToolkit admin UI; draft entities in Cosmos; publish / scheduled publish to `PublishedConfiguration`; public `GET /api/config/v1` (published only)
3. **Azure Cosmos DB** - draft entities and published snapshots
4. **price-configurator (Zeus React)** - fetch published config at init; `getPrice / utils.js`; GraphQL quote submit to price-configurator-api (unchanged)

```
Editors | v price-configurator-back-office | v Azure Cosmos DB | v  
price-configurator (Zeus React) ---> price-configurator-api (quote submit)
```

3.2 Boundaries (what each system owns)

System	Owns	Does not own
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price-configurator-back-office	Catalog structure, prices, CMS copy (phase 3+), images (Cloudinary IDs), publish lifecycle, public read API	Live total calculation, customer quote storage
price-configurator	UX, selection state, <code>getPrice</code> logic, email/PDF client calls	Authoritative price tables (after migration)
price-configurator-api	Quote persistence, email, Marketo, Cosmos customer artifacts	Product catalog (unchanged role)

3.3 Reference stack (kitchen-quiz-backoffice)

Layer	Technology
Runtime	.NET 8, ASP.NET Core
CMS	Nobia.CmsToolkit 1.26.x
Database	Cosmos DB via EF Core Cosmos provider
Auth	Cookie + Azure AD (Auth Scripts API)
Images	Cloudinary via CmsToolkit <code>CloudinaryImage</code>
API docs	NSwag (non-production)

Deploy	Nobia Digital platform (infra/template.yaml, Azure Pipelines)
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New service name: price-configurator-back-office (repo and platform service id aligned with kitchen-quiz naming).

4. Key design decisions

#	Decision	Rationale
D1	Normalized draft entities + published snapshot on publish	Atomic go-live; supports schedule; avoids half-published catalog on public API
D2	Frontend-only price calculation	Reuses proven utils.js; smaller API; matches current UX latency
D3	Public API shape matches today's data export	Minimal React refactor; same getPrice field names (prices, prebuiltPrices, handlelessPrices, etc.)
D4	Draft + Publish + Scheduled publish (v1)	Business control; kitchen-quiz does not implement this - must be built explicitly

D5	Preview URLs deferred (post-v1)	Reduce v1 scope; editors validate in CMS until preview phase
D6	Stable keys on all cards/ranges (rangeKey, cardKey)	Maps CMS rows to frontend selection and price logic
D7	API versioned (/api/config/v1)	Safe evolution during Magnet pilot

5. Content model (CMS entities)

Editors work with **normalized** entities. The public site never reads draft rows directly.

5.1 Entity overview

Entity	Cardinality	Purpose
ConfiguratorSettings	Singular per brand + language	Section order, currency, months-to-pay, header height, global flags
ConfiguratorSection	Many	Wizard step: units, range, worktop, installation, ...

ConfiguratorCard	Many	One selectable option; links to section; Cloudinary image
RangePriceSet	Many (Magnet-heavy)	prices, prebuiltPrices, handlelessPrices per unit tier
PriceTable	Few per brand	Shared matrices: worktops, installation bands, appliances, sinks, flooring
CardOption	Many	Handles, sink types, appliance tiers with nested prices

5.2 Published snapshot

On **Publish** (manual or scheduled):

1. Validate required fields (images, price array lengths vs unit tiers, references).
2. Run **snapshot builder**: draft entities -> single PublishedConfiguration document.
3. Store with version, publishedAt, brand, language.

4. Public GET /api/config/v1 reads **only** this snapshot.

6. Public API

6.1 Endpoint

```
GET /api/config/v1?brand={brand}&language={language}
```

Response	Condition
200 + JSON body	Published snapshot exists
404	No published config for brand/language

Headers (recommended): ETag or body field `version` for cache busting after publish.

6.2 Response contract

JSON MUST be consumable by the existing frontend pipeline:

- Equivalent to what `src/brands/data.js` produces after `createCloudinaryUrl` mapping (or document whether image URLs are built client-side from `PublicIds`).
- Preserve fields required by `getPrice` in `src/components/cards/utils.js`.

6.3 Out of scope for v1

- `POST /api/estimate` (server-side totals)
- Preview token endpoints (phase 4)

7. Publish & schedule workflow

State	Visible to
Draft	CMS editors only
Scheduled	<code>ScheduledPublishAt</code> set; worker publishes at that time
Published	<code>PublishedConfiguration</code> snapshot; public API

7.1 Manual publish

1. Editor clicks **Publish** in admin UI.
2. Validation runs on draft aggregate.
3. Snapshot written; `version` incremented.
4. CDN/API cache invalidated or short TTL (target: live site within minutes).

7.2 Scheduled publish

Same snapshot build at `ScheduledPublishAt` via background job (Azure Function, `IHostedService`, or platform scheduler - align with Nobia standards).

7.3 Frontend caching

- Fetch config on app load (and optional refresh interval).
- Use `version` query param or `ETag` after publish.
- Avoid long-lived cache that hides price updates for hours.

8. Frontend migration (price-configurator)

Today	Target
Static import from <code>src/brands/{brand}/index.js</code>	<code>fetch(CONFIGURATOR_API_URL + '/api/config/v1?...')</code> at init
<code>prices.js</code> per brand	Removed when brand is API-driven
<code>messages.json</code>	Retained until phase 3; then CMS

utils.js	Unchanged responsibility; verify mapped API fields
GraphQL submit	Unchanged

8.1 Environment

Add Zeus/settings key, e.g. configuratorApiUrl, per environment (dev/test/stage/prod).

8.2 Error handling

Production MUST NOT silently fall back to bundled src/brands if API fails (configurable dev-only fallback optional).

9. Phased delivery

Phase P0 - Platform foundation

Goal: Deployable empty backoffice matching kitchen-quiz patterns.

Deliverable	Details
New solution price-configurator-back-office	.NET 8 web, csproj, sln, nuget.config
CmsToolkit + Cosmos	Connection string, container, brand registration
Auth	Azure AD groups per brand (mirror Constants pattern)
Health + infra	infra/template.yaml, Azure Pipelines
Admin UI shell	Entities registered; no public data yet

Exit criteria: Service deploys to dev; admins can log in; health check passes.

Phase P1 - Magnet pilot (catalog + prices + publish)

Goal: Magnet live on API; frontend uses published config only for Magnet.

Deliverable	Details
CMS models	All Magnet sections/cards/price matrices
Import tool	One-time migration from <code>src/brands/magnet/*</code> + <code>prices.js</code>
Snapshot builder + publish	Manual publish works
Scheduled publish	Job promotes scheduled drafts
<code>GET /api/config/v1</code>	Magnet + en-GB (and required languages)
Frontend	Magnet brand loads from API; loading/error states
Validation	Publish blocked on missing images/prices

Exit criteria: Editor changes range price -> publish -> Magnet site shows new price without React deploy.

Still in repo for Magnet: `messages.json` copy (phase 3).

Phase P2 - Remaining brands

Goal: All six brands on published API; remove hardcoded brand folders as each goes live.

Order (suggested)	Relative effort
invita	Lower (~300 lines index)
sigdal, norema, novart	Medium
marbodal	High (similar to Magnet)
magnet	Done in P1

Per brand: import -> CMS -> publish -> frontend switch -> delete `src/brands/{brand}` + `prices.js`.

Exit criteria: No brand relies on bundled `src/brands` for catalog/prices in production.

Phase P3 - Copy & translations in CMS

Goal: Retire `config/{brand}/messages.json`; tooltips and section copy in CmsToolkit with `[Translated]`.

Deliverable	Details
ConfiguratorMessages or fields on Settings/Sections	Per kitchen-quiz translation pattern
API extension	v1 payload includes message keys or inline copy
Frontend	Remove messages.json imports per brand

Exit criteria: Copy changes do not require price-configurator repo deploy.

Phase P4 - Preview

Goal: Editors preview draft config before publish.

Deliverable	Details
Authenticated preview API or tokenized URL	Draft snapshot, not public cache
Optional Zeus preview mode	Query param or separate host

Exit criteria: Marketing can sign off without publishing to production.

Phase summary table

Phase	Focus	Public API	Frontend
P0	Skeleton + deploy	-	Unchanged
P1	Magnet + publish	v1 Magnet	Magnet -> API

P2	All brands	v1 all brands	All -> API
P3	Copy in CMS	v1 + messages	No messages.json
P4	Preview	preview endpoint	Preview mode

10. Migration strategy

10.1 Data import

1. Parse existing `index.js` + `prices.js` into intermediate JSON (Node script or C# tool).
2. Upsert draft Cosmos entities with stable keys.
3. Review in CMS admin.
4. **Publish** to create first `PublishedConfiguration`.

10.2 Rollback

- Keep previous `PublishedConfiguration` version N-1; republish to rollback.

- Frontend can pin `?version=` in emergencies (optional operational runbook).

10.3 Parallel run (Magnet pilot)

- Feature flag: `USE_CONFIG_API=true` for Magnet in dev/stage before prod cutover.

11. Security & operations

Area	Approach
Admin access	Azure AD groups per brand (same as kitchen-quiz <code>Startup.cs</code> pattern)
Public API	Read-only; CORS for Zeus origins; no auth on GET config (published data only)
Secrets	Cosmos, Cloudinary, Auth Scripts API via platform secrets
Observability	Application Insights, Serilog (<code>Nobia.Backend.*</code> packages)

12. Risks & mitigations

Risk	Mitigation
API shape drift breaks getPrice	Contract tests: snapshot JSON vs golden fixture from current Magnet export
Large Magnet model hard to edit	Section/card listing pages in CMS; not one giant form
CmsToolkit lacks native schedule/publish	Explicit PublishedConfiguration entity + hosted scheduler
Editors publish broken config	Publish-time validation (images, price dimensions)
Stale CDN cache after publish	Short TTL + version in URL
Quote submit price mismatch	Document that submit stores client totals; server re-validation future hardening

13. Success criteria (program level)

1. Business users edit kitchen prices in backoffice without developer involvement.
2. Published changes appear on the live configurator within agreed SLA (minutes, not deploy cycle).
3. All six brands served from Cosmos published snapshots.
4. price-configurator repo contains UI/code only - no authoritative prices.js.
5. Draft/scheduled publish prevents accidental live price changes.

14. References

Resource	Path
Price configurator README	<code>price-configurator/README.md</code>
Brand data (example)	<code>price-configurator/src/brands/magnet/</code>
Price logic	<code>price-configurator/src/components/cards/uti</code>
Kitchen quiz stack doc	<code>kitchen-quiz-backoffice/.planning/codebase/</code>
Kitchen quiz architecture	<code>kitchen-quiz-backoffice/.planning/codebase/</code>
Kitchen quiz API pattern	<code>GET /api/settings/v2 in ApiController.cs</code>

15. Glossary

Term	Meaning
GAGP	Magnet's name for Price Configurator
Published snapshot	Immutable JSON document served to the public API
Stable key	Machine id (ascoli, worktop-laminate) surviving CMS edits to display title
Zeus	Nobia frontend hosting/tooling for React apps

End of strategy document